# ANALYTICAL DATASET FOR DEMAND FORECASTING IN A SCHOOL RESTAURANT: DEVELOPMENT THROUGH DESIGN SCIENCE RESEARCH, EXPLORATORY ANALYSIS, AND FEATURE ENGINEERING


## ABSTRACT

Meal planning in school restaurants depends on demand estimates that are subject to operational uncertainties , such as no-shows despite reservations, consumption without prior reservation, recording failures, and variations associated with the academic calendar. This article presents the development and exploratory evaluation of an analytical dataset to support predictive demand models for a school restaurant, based on the Design Science Research approach. The data flow was organized into Bronze, Silver, and Gold layers, integrating reservations, consumption records, menus, academic calendar, and weather data from 2023 to 2025. As a result, the Gold layer consolidated 548 records suitable for modeling and 140 predictive attributes, with an explicit separation between observed consumption, estimated consumption, variables used for modeling, and variables eligible for prediction. The exploratory analysis revealed weekly patterns, seasonal regimes, menu effects, and the centrality of reservations as an explanatory signal of demand. An exploratory semantic representation of the menu was also prepared using BERT and PCA, preserving approximately 95.16% of the variance in 44 components. The artifact contributes to reducing opacity in data preparation, mitigating the risk of temporal leakage, and providing a traceable dataset for future model comparisons aimed at production planning and food waste reduction.



**Keywords: demand forecasting; school restaurant; food waste; Design Science Research; exploratory data analysis.**


## 1. INTRODUCTION

Food waste in collective food services is an operational, economic, and socio-environmental problem. In school restaurants, the daily decision about how many meals to produce must balance two risks: preparing less than the effective demand and compromising service, or preparing more than demand and increasing losses. This issue becomes more critical when planning depends on reservations, manual records, and variations in the academic calendar, such as holidays, recesses, and the beginning and end of academic periods.

Recent literature shows that machine learning models can reduce waste when they replace purely intuitive estimates with forecasts based on historical data, calendar information, and operational context. Rodrigues et al. (2024), for example, analyzed collective food services

and demonstrated that demand forecasts can support the reduction of wasted meals when incorporated into planning. Similarly, Turker (2025) discusses university dining facilities as environments in which calendar, menu, weather, and attendance behavior data can support sustainable decisions.

However, the usefulness of predictive models does not depend only on the chosen algorithm. Before comparing families of predictive models, such as statistical time series, tree-based methods, linear models, neural networks, or hybrid approaches, it is necessary to build a traceable and coherent data artifact capable of distinguishing what was observed, what was estimated, and what should be used for modeling. Without this separation, the model may learn patterns fabricated by inadequate data treatment, producing optimistic metrics and fragile decisions.

This study treats the data flow as the central research artifact, guided by Design Science Research (DSR). This choice is consistent with Hevner et al. (2004), Peffers et al. (2007), and Gregor and Hevner (2013), because the objective is not only to describe a phenomenon but to design, demonstrate, and evaluate an informational solution to a practical problem. The proposed artifact transforms heterogeneous datasets into a documented daily analytical dataset prepared for future modeling stages.

The general objective of this article is to develop and conduct an exploratory evaluation of a Gold analytical dataset for future meal demand forecasting in a school restaurant. In this study, the Bronze, Silver, and Gold layers are understood, respectively, as traceable raw data, consolidated and qualified data, and analytical data prepared for modeling. Thus, the Gold dataset represents the final artifact of this research stage, while the previous layers constitute the processes required for its construction.

To achieve this general objective, five sequential specific objectives were defined. The first consists of structuring the ingestion of reservation, consumption, menu, calendar, and weather sources into the Bronze layer, preserving traceability. The second consists of consolidating a daily Silver dataset, distinguishing observed, estimated, and non-applicable data. The third seeks to conduct an exploratory analysis to evaluate temporal patterns, seasonality, variability, and menu effects. The fourth consists of generating Gold datasets with temporal, operational, weather, and menu-related attributes. The fifth consists of evaluating the feasibility of an exploratory semantic menu block based on BERT and PCA, to be validated in future models.

## 2. THEORETICAL BACKGROUND

### 2.1 Demand forecasting and waste in food services

Collective food services operate under daily uncertainty. In school and university restaurants, demand may vary by day of the week, academic period, days preceding holidays, exams, weather conditions, and menu attractiveness. The problem is close to what Malefors (2022) describes in the public collective food sector: the quantity prepared in relation to the actual number of users served is a central factor in waste generation.

Rodrigues et al. (2024) reinforce that short-term demand forecasting is a concrete strategy to reduce both overproduction and unmet demand. Their study is especially relevant to this work because it uses real food service data and compares machine learning models with traditional estimation approaches. This supports the decision to prepare a daily, multivariate, and traceable database before advancing to predictive models.

Chaudhari et al. (2025) also highlight the importance of historical data, analytical modules, and algorithms such as XGBoost, LightGBM, and SARIMA in forecasting systems for food supply chains. Although the context of the present study is a school restaurant, the analytical logic is similar: useful forecasting depends on reliable history, consistent explanatory variables, and the ability to interpret trends before operational decision-making.

### 2.2 Design Science Research and analytical dataset construction

DSR is appropriate when research seeks to build and evaluate an artifact capable of solving an identified problem in a real context. Hevner et al. (2004) present design research as a paradigm aimed at creating artifacts that expand human and organizational capabilities. Peffers et al. (2007) structure this process into stages such as problem identification, objective definition, design and development, demonstration, evaluation, and communication.

In this article, the artifact is not yet the final predictive model, but the analytical dataset and preparation flow that make such a model possible. This distinction is important because, in data science projects, result robustness depends on decisions made before training: date standardization, removal of duplicate records, treatment of missing values, category encoding, creation of temporal lags, and preservation of quality markers. Tripathi et al. (2021) advocate more robust data science processes precisely because they recognize that data failures and modeling failures are interdependent.

The adoption of Bronze, Silver, and Gold layers follows this concern. Bronze preserves ingested data with minimal transformation, maintaining traceability. Silver consolidates daily semantics and separates observations, estimates, and modeling criteria. Gold creates predictive

attributes already oriented toward use by different families of models. This architecture avoids mixing ingestion, cleaning, imputation, and modeling into a single opaque step.

### 2.3 Menu, natural language, and feature engineering

The menu is a relevant contextual variable because users' choices are not determined only by calendar and reservation. Kim et al. (2023) explore the use of unstructured data and textual representations, such as Word2Vec, to forecast cafeteria customers. This type of evidence does not eliminate the need for interpretable categories, but it supports the methodological hypothesis that dish descriptions may carry additional signals of food preference.

Transformer-based models expanded the ability to represent text through contextual embeddings. Vaswani et al. (2017) introduced the Transformer architecture, and Devlin et al. (2019) proposed BERT, a bidirectional model capable of generating dense text representations. In this study, BERT is not assumed to be superior to manual categories; it is treated as an exploratory alternative to represent semantic similarities that can later be tested against models based only on protein, preparation, and style.

For example, two dishes manually classified under the same protein category may carry distinct preference signals, such as "grilled chicken" and "chicken fricassee." Although both belong to the poultry group, their textual descriptions indicate different preparation methods, dish composition, and sensory characteristics. Embedding representations seek to preserve part of these semantic relationships in numerical format, allowing future testing of whether such information adds value to traditional categorical attributes.

The representations generated by BERT assign an extensive numerical vector to each menu description. Considering the limited number of modelable records, using all these dimensions could increase model complexity and raise the risk of overfitting.

For this reason, Principal Component Analysis (PCA) was applied, a dimensionality reduction technique that transforms correlated variables into a smaller set of components while preserving most of the data variation (Jolliffe & Cadima, 2016). Thus, BERT and PCA are treated as an exploratory strategy to compress textual menu information, not as evidence of causality or as prior proof of predictive gain.

## 3. METHODOLOGY

### 3.1 Research design

The research follows an applied, exploratory approach guided by DSR. The practical problem consists of preparing a reliable dataset for meal demand forecasting in a school restaurant. The developed artifact is a data flow organized in computational notebooks, responsible for ingesting heterogeneous sources, consolidating daily granularity, documenting quality, and generating predictive attributes for future machine learning models.

The temporal scope covers the period from January 1, 2023, to December 31, 2025, totaling 1,096 calendar days. The sources include individual reservations, individual consumption records, menus, academic calendar, and weather. Ingestion was separated by domain, avoiding premature coupling between variables of different natures. This separation also facilitates auditing, re-execution, and localized correction when a source presents inconsistencies.

### 3.2 Ingestion and Bronze layer

The Bronze layer was built through five ingestion computational notebooks: reservations, consumption records, menus, academic calendar, and weather variables. This division was adopted because each source has its own rules for reading, duplicate handling, dates, mandatory columns, and operational interpretation.

At the end of ingestion, 84,453 reservation records, 66,869 consumption records, 1,095 menu records, 1,096 calendar observations, and 1,096 weather observations were preserved. In the Bronze layer, data were persisted in two complementary formats: Parquet, due to its storage efficiency, preservation of data types, and ease of reading in computational notebooks; and CSV, because it favors manual inspection, processing transparency, and auditing of raw records. This decision meets the traceability requirement: any subsequent transformation can be compared with the ingested source.

### 3.3 Silver consolidation

The Silver layer consolidates the sources at daily granularity. This step is decisive because demand forecasting models will be trained by day, not by individual. Therefore, individual reservations and consumption records were aggregated into daily metrics, while calendar, weather, and menu data were aligned by date. The result was a master dataset with 1,096 rows and 77 columns.

A central concern was to differentiate observed days, estimated days, and days not applicable for modeling. Instead of silently replacing missing values with imputed values, Silver records modeling criteria and quality markers. This choice reduces the risk of transforming collection failure into real demand behavior. It also allows future models to use only modelable days or to test scenarios with different tolerance levels.

Missing data treatment was conducted conservatively and traceably. For weather variables, a simple filling rule was planned in case any day lacked records, using nearby days as a reference. In the final dataset, however, this routine had no relevant impact, because weather records did not present significant gaps.

For the menu, imputation was restricted to school days without dish information and used the mode by day of the week, that is, the most frequent category observed on that same weekday across the historical series. When there was insufficient information in this grouping, the global mode of the dataset was used as an alternative.

Missing reservations and consumption records, or very low consumption values classified as suspicious, were estimated using a linear regression model fitted only with days considered valid, using calendar, reservation, weather, and menu variables. When the amount of valid data was insufficient for fitting, the temporal median was used as a conservative alternative. In all cases, original observed values were preserved in separate columns and imputed records were marked for auditing.

Thus, Silver does not erase data uncertainty. It creates three complementary views: observed values, estimated values, and values intended for modeling. Modeling columns are used to form complete series, while imputation markers and applied criteria enable later auditing. This decision is important because it prevents an estimated value from being confused with a direct empirical observation.

### 3.4 Gold engineering

The Gold layer transforms Silver into a machine learning matrix. Calendar variables, consumption and reservation lags, moving averages, weather attributes, categorical menu encodings, and relevant interactions were created. After the filters needed to avoid initial rows without sufficient temporal history, Gold resulted in 548 records and 140 attributes.

The derived variables were built from information available before the meal, reducing the risk of temporal leakage. Consumption and reservation lags, moving averages calculated with temporal shifting, calendar indicators, cyclical transformations of weekday and month, weather attributes, and menu indicator variables were generated. This strategy makes it possible to represent recent demand memory, institutional seasonality, and operational context.

Variables that would carry direct information from the outcome itself were also excluded from the list of predictive inputs, such as observed consumption, estimated consumption, modeling consumption, and audit markers. These fields remain in the dataset for traceability, but should not be used as direct model inputs. Thus, Gold differentiates the target variable, quality metadata, and attributes effectively eligible for prediction.

This stage also separates families of attributes. Linear models tend to benefit from scaled variables, controlled encodings, and lower collinearity; tree-based models better handle nonlinearities, interactions, and sparse indicator variables; temporal models depend on lags and moving averages. Therefore, feature engineering was documented not as a single universal set, but as a catalog of attributes that can be selected according to the algorithm.

### 3.5 Exploratory semantic representation with BERT and PCA

In addition to categorical menu variables, an exploratory textual representation with BERT was prepared. The input text was formed by combining the main protein and preparation attributes, generating a short description of each day's menu. Then, BERT transformed these descriptions into numerical vectors, enabling computational representation of textual similarities between menus.

Because these vectors have many dimensions for a dataset with 548 modelable records, Principal Component Analysis (PCA) was applied to compress the textual representation. The procedure reduced semantic information to 44 components, preserving about 95% of the accumulated variance. This stage was kept as an exploratory and complementary block, without replacing the interpretable protein, preparation, and dish style variables.

## 4. RESULTS AND DISCUSSION

### 4.1 Formation of the data artifact

Before interpreting consumption patterns, it is necessary to understand how the analytical dataset was formed. Figure 1 presents the artifact preparation funnel, indicating the transition from the complete calendar to the subsets effectively useful for analysis and modeling.

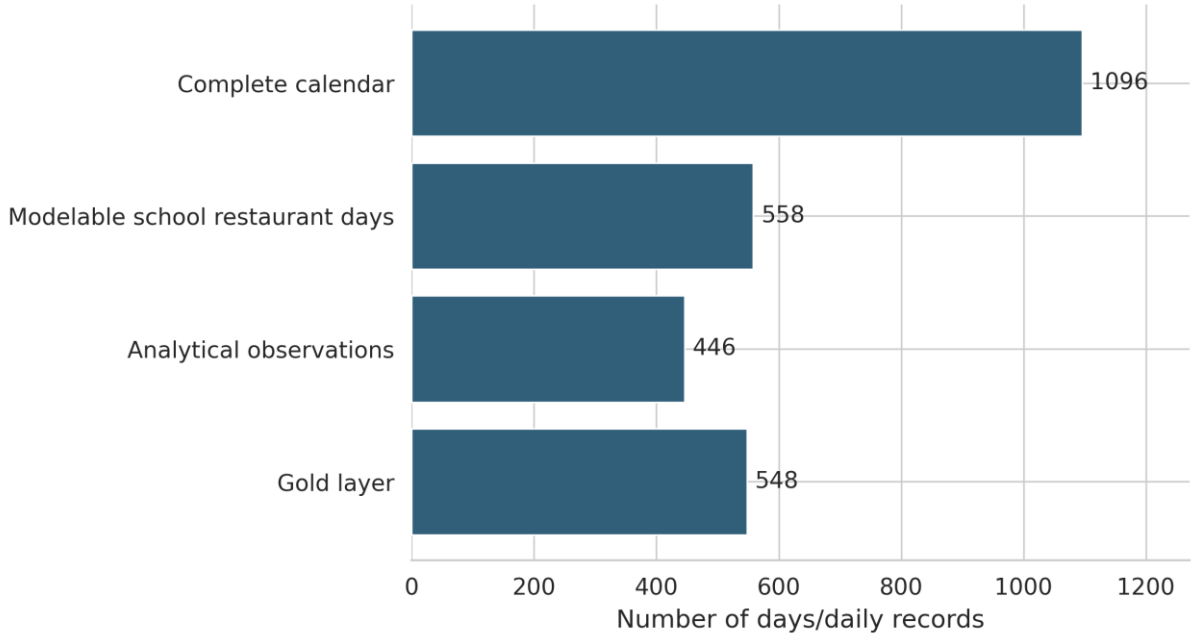



Figure 1. Data artifact preparation funnel. Source: authors' own elaboration.

Figure 1 summarizes the transition from the complete calendar to the analytical dataset. The data flow starts from 1,096 days, identifies 558 modelable days for the school restaurant, maintains 446 analytical observations with sufficient observed data, and reaches 548 records in the Gold layer after the creation of temporal variables. The apparent difference between analytical observations and Gold occurs because Gold incorporates temporal structure and can retain modelable records with explicit criteria.

This funnel is important because it prevents a naive reading of the dataset. Not every calendar day represents a useful operating day for the restaurant, and not every operating day has observed consumption with sufficient quality. The visualization demonstrates the need to distinguish calendar availability, operational applicability, and readiness for modeling, which is aligned with Tripathi et al.'s (2021) concern with the robustness of data-driven processes.

### 4.2 Temporal dynamics of reservations and consumption

After delimiting the available data volume, the next step consists of observing how reservations and consumption behave over time. Figure 2 compares these two series because the difference between consumption intention and effective consumption is one of the main sources of operational uncertainty in the school restaurant.

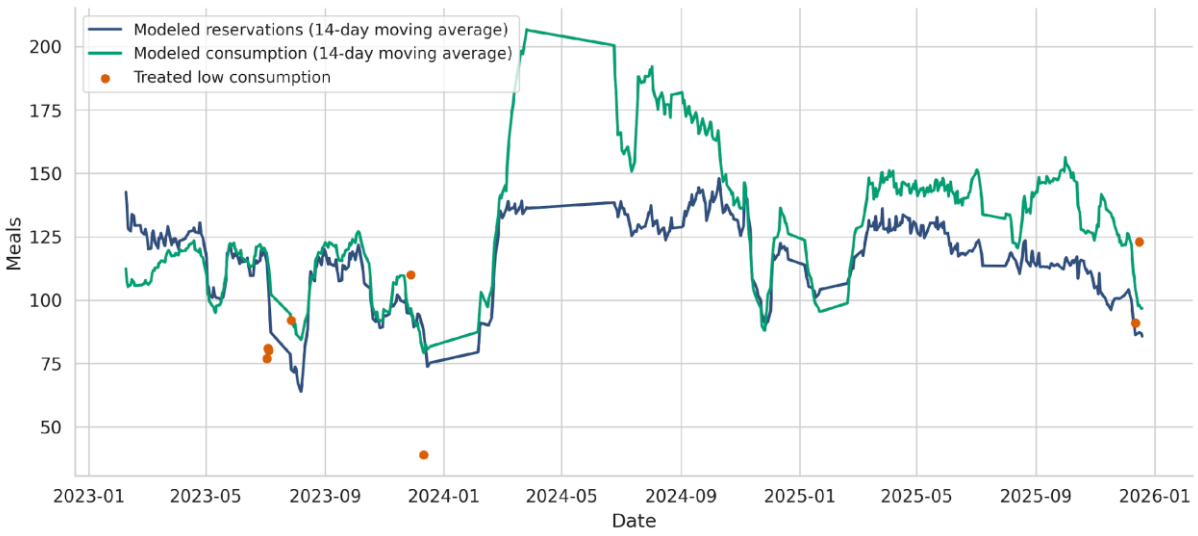



Figure 2. Temporal evolution of reservations and modeled consumption in the school restaurant. Source: authors' own elaboration.

Figure 2 presents the temporal evolution of reservations and modeled consumption, smoothed by a 14-day moving average. This procedure calculates, for each date, the average of values observed in previous days, reducing point fluctuations and facilitating visualization of the general demand trend.

The distance between the lines indicates that reservation is a relevant signal, but it is not automatically equivalent to effective consumption. This behavior is compatible with the literature on attendance forecasting in food services, in which planned counts need to be adjusted by the users' actual presence (Malefors, 2022; Rodrigues et al., 2024).

The points highlighted as suspicious low consumption represent records that require methodological attention. Because consumption is manually recorded, very low values on operating days may reflect collection failures, and not necessarily real low demand. Therefore, the artifact does not erase these records; it flags them and creates modeling criteria, allowing future models to test the impact of including them or treating them conservatively.

### 4.3 Weekly demand profile

In addition to the aggregated temporal evolution, demand must be observed within the weekly operating routine. Figure 3 focuses the analysis from Monday to Friday, because these are the days that represent regular student meal service in the studied context.

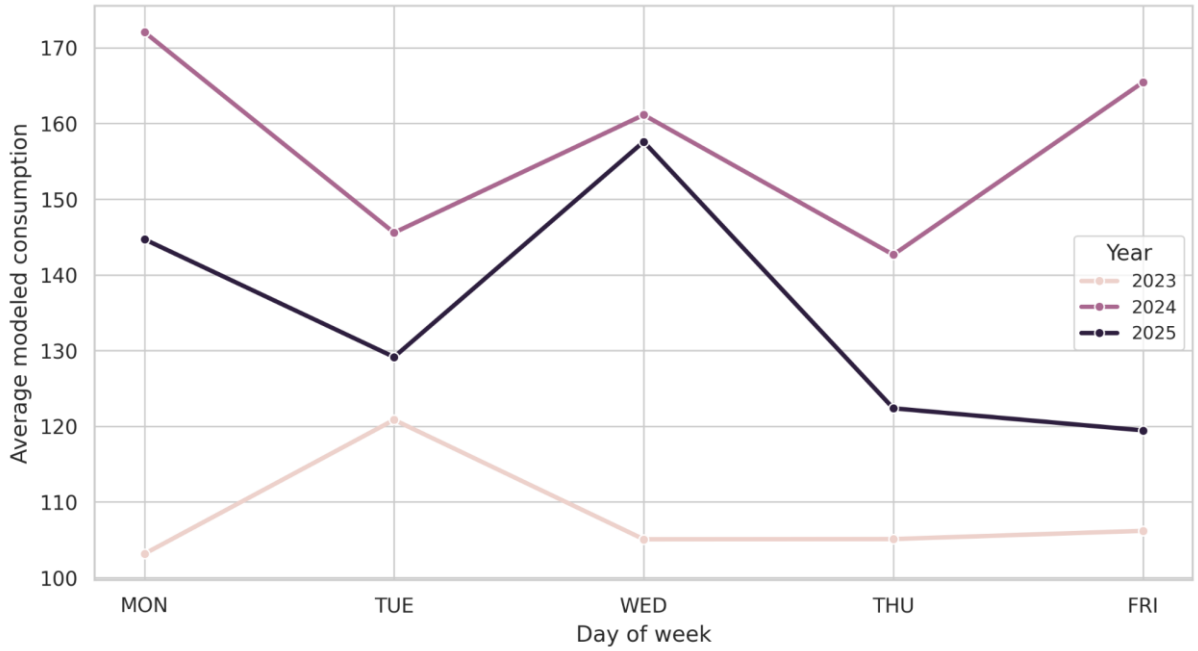



Figure 3. Average weekly demand profile, from Monday to Friday. Source: authors' own elaboration.

Figure 3 shows that average consumption is not uniformly distributed throughout the week. There are differences between the analyzed years and between weekdays, indicating that the position of the day in the weekly calendar carries relevant information for demand forecasting.

This result justifies the creation of calendar variables in the Gold dataset. To represent cyclical patterns, such as weekday and month, sine and cosine transformations were used, allowing the model to recognize proximity between recurring periods. The literature on forecasting in collective food services also points to calendar and seasonality as relevant factors, especially when operations depend on school routines, holidays, and academic periods (Rodrigues et al., 2024; Turker, 2025).

### 4.4 Monthly intensity and operational seasonality

Because the week alone does not explain demand variation, it is also necessary to observe whether certain months concentrate higher or lower consumption. For this purpose, Figure 4 presents a heatmap that crosses month and weekday, allowing demand intensity to be visualized in a single representation.

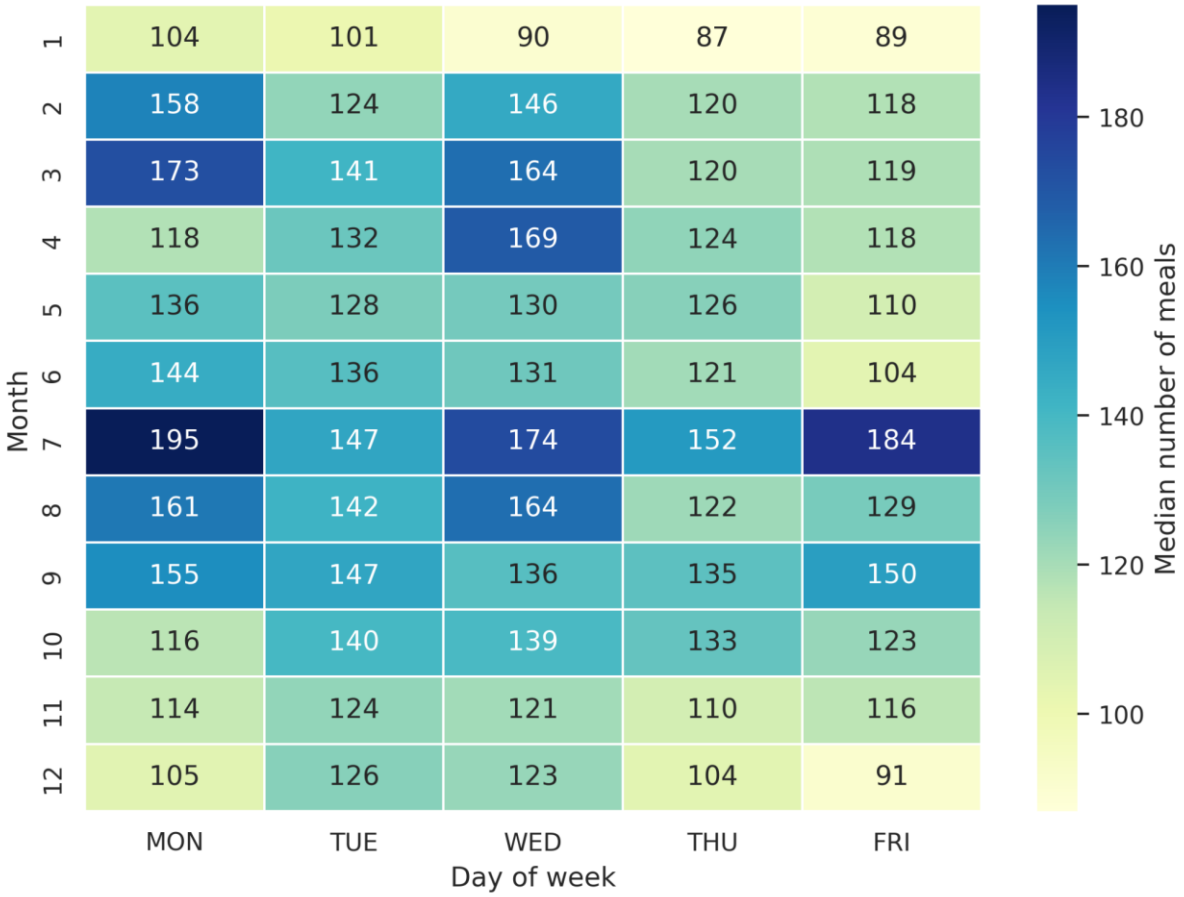



Figure 4. Demand intensity by month and weekday. Source: authors' own elaboration.

Figure 4 indicates that demand varies not only between weekdays, but also between months of the year. Periods of greater intensity are observed, such as months with more regular academic operation, as well as periods of lower demand, possibly associated with vacations, recesses, or academic transitions.

This pattern reinforces that demand should not be modeled only as a mechanical continuation of the previous day. There are seasonal and institutional components associated with the academic calendar, and these components need to be explicit in the Gold dataset through variables such as month, year, holiday, school day, academic events, and operation indicators.

### 4.5 Variability of observed consumption

After evaluating average patterns, it is important to examine the dispersion of observed data. Figure 5 presents a boxplot to compare the distribution of consumption between weekdays, allowing the observation of median, amplitude, concentration of values, and extreme records.

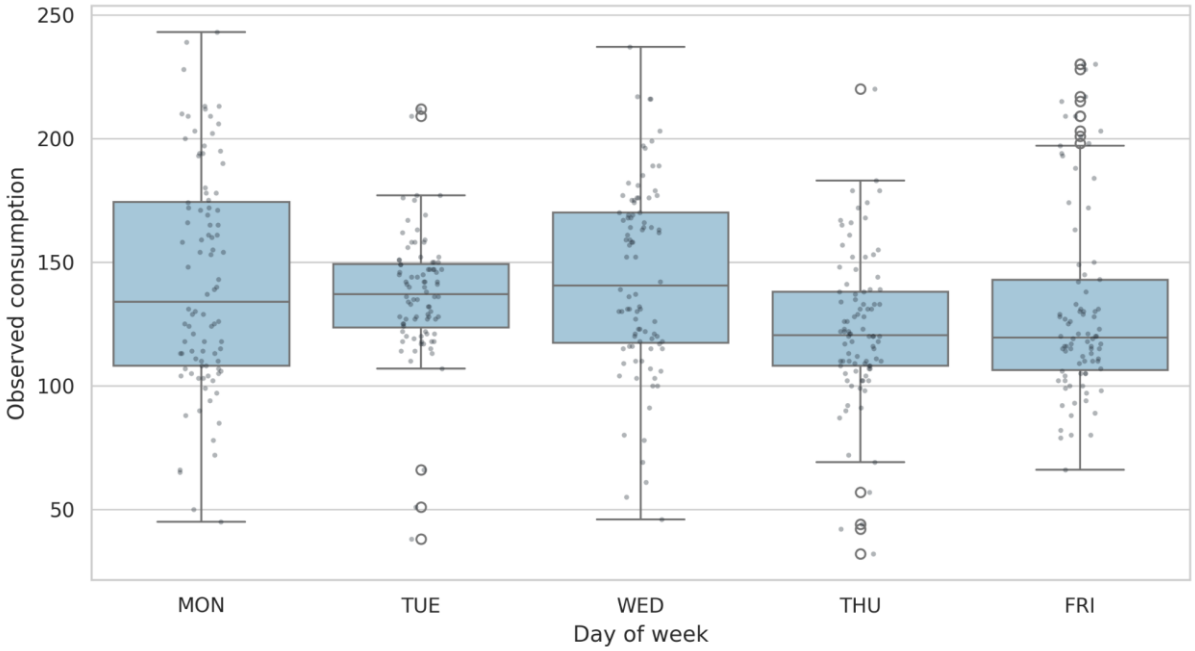



Figure 5. Dispersion of observed consumption by weekday. Source: authors' own elaboration.

Figure 5 shows that consumption variability is not the same on all weekdays. Some days present greater amplitude and a greater presence of extreme values, indicating that demand may be affected by punctual events, routine changes, or atypical operational records.

This result reinforces that it is not enough to predict a general average consumption. Future models will need to deal with variability, extreme values, and regime changes. For the current stage, the main contribution is to diagnose this heterogeneity before modeling, avoiding algorithm selection without understanding the empirical distribution of the target variable.

### 4.6 Effect of protein type on consumption

The menu was initially analyzed by main protein, because this category provides a simple and interpretable reading of food composition. Figure 6 presents differences in median demand between protein groups, functioning as an initial diagnosis of the possible menu effect on consumption.

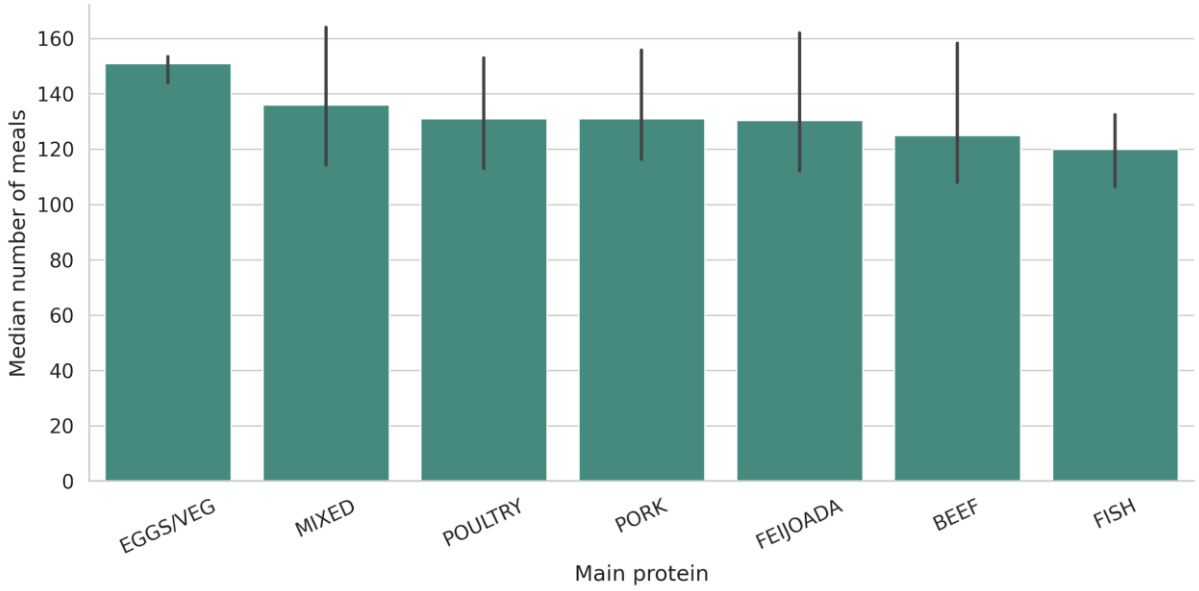



Figure 6. Median demand by main menu protein. Source: authors' own elaboration.

In Figure 6, differences in median demand between protein groups are presented, but this reading must be interpreted with caution. The variation bars indicate that there is dispersion within each group, which prevents claiming direct causality between protein and consumption. Even so, the result reinforces that menu attributes may carry useful information for demand forecasting, in dialogue with Kim et al. (2023), who treat textual menu information as a potential source of forecasting in cafeteria environments.

The refinement of the menu computational notebook reduced the residual protein category to only two cases classified as "Other," which strengthens the quality of this attribute block. Still, categories such as "no operation" and days without a menu must be interpreted carefully, because they represent operational context and not food preference. This distinction prevents the model from confusing absence of operation with rejection of a given dish.

### 4.7 Interaction between protein and preparation

Protein analysis alone does not capture all menu information. For this reason, Figure 7 crosses protein and preparation, allowing verification of whether certain food combinations appear associated with differences in median demand.

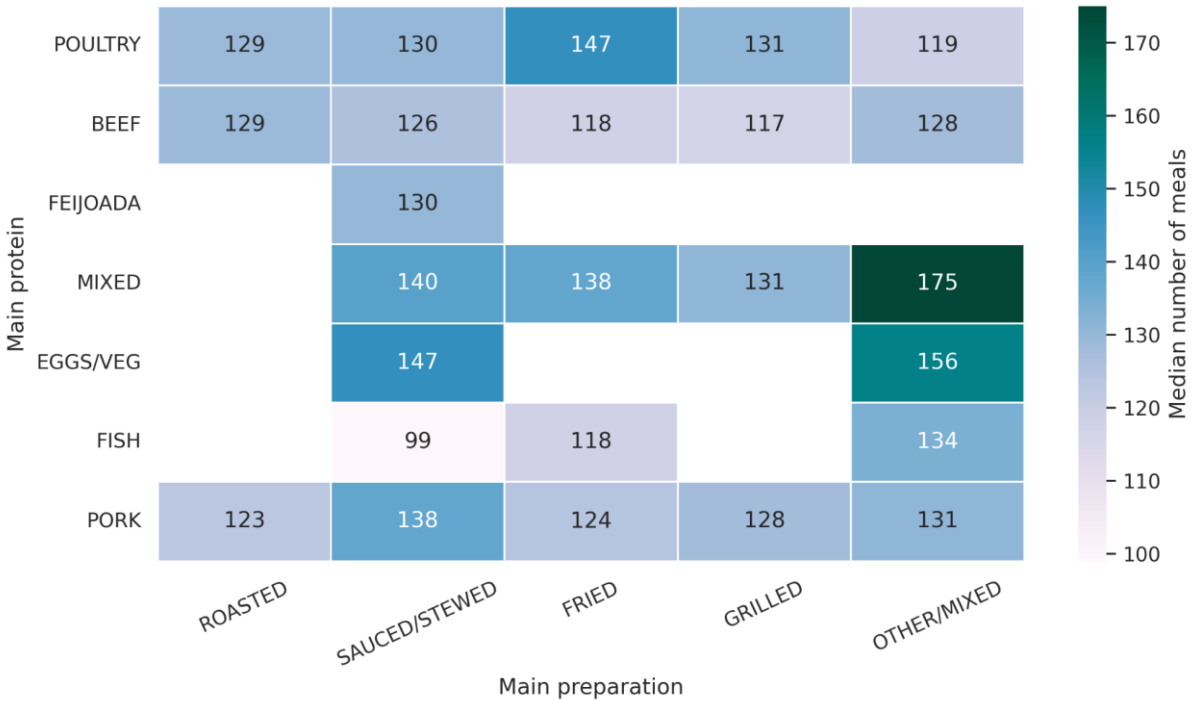



Figure 7. Median demand by combination of protein and preparation. Source: authors' own elaboration.

Figure 7 indicates that demand may vary according to the combination between main protein and preparation method. For example, similar protein categories may present different medians when associated with preparations such as stewed in sauce, fried, grilled, or roasted. This reading does not imply direct causality, but suggests that menu structure contains relevant contextual signals.

This visualization justifies the creation of categorical menu attributes and possible interactions between food families. From the perspective of feature engineering, it shows why the menu was treated at three levels: main protein, main preparation, and dish style. Style maintains a larger residual category because not every dish has a specific culinary identity, but this is acceptable when the variable is complementary and does not replace protein and preparation.

### 4.8 Exploratory attribute selection

With the Gold dataset built, it became possible to conduct a first analysis of attribute relevance. Figure 8 summarizes this analysis through the importances estimated by ExtraTrees, a method based on extremely randomized trees (Geurts et al., 2006). The use of the method at this stage is exploratory and seeks to guide future comparisons between sets of variables.

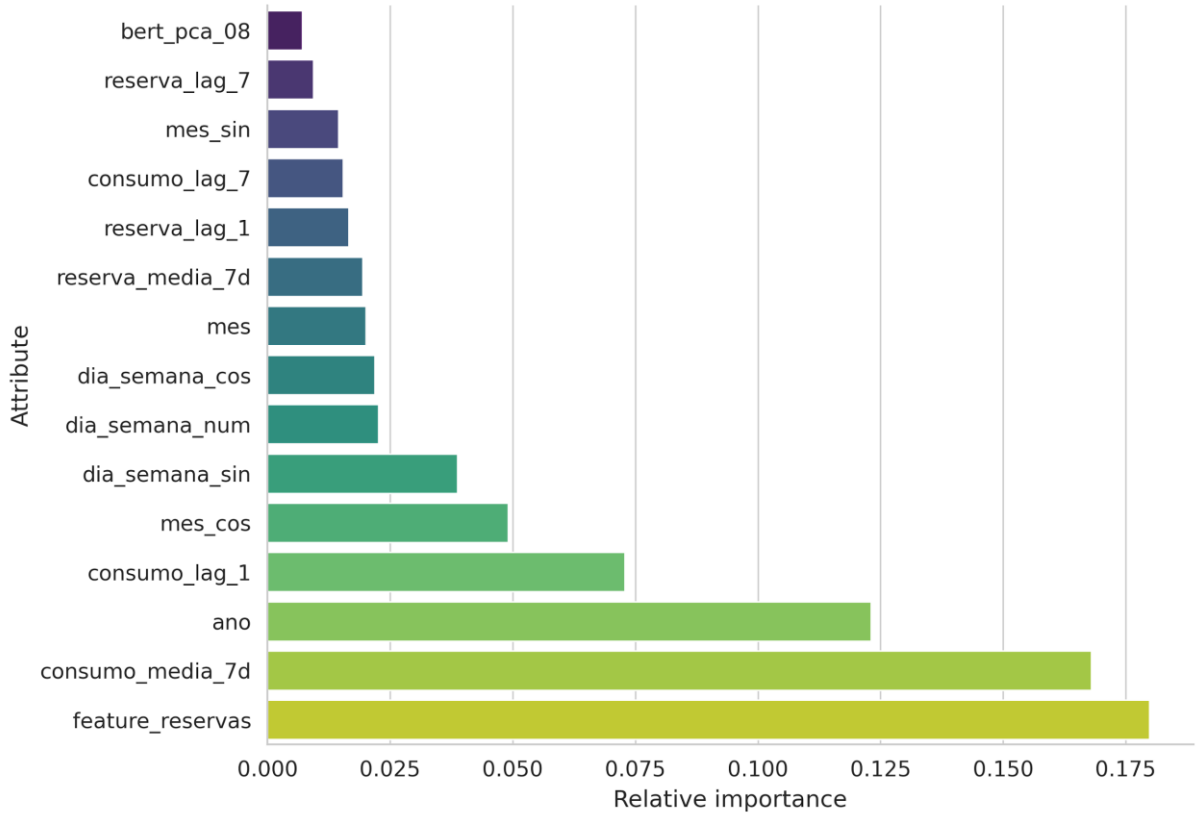



Figure 8. Most relevant predictors according to ExtraTrees. Source: authors' own elaboration.

Figure 8 highlights the centrality of reservations as a predictive signal, while also emphasizing moving averages, temporal lags, calendar, and contextual variables. This reading is coherent with the demand forecasting literature, which values recent history and operational variables to anticipate consumption (Chaudhari et al., 2025; Rodrigues et al., 2024).

It is important to treat this figure as supporting analysis, not as a final conclusion of causality. Tree-based importances may favor variables with greater separation power within the sample and depend on preprocessing choices. Even so, the ranking is useful to guide the next modeling stage, especially when comparing linear, temporal, categorical, and semantic attribute sets.

### 4.9 Semantic representation of the menu

To complement the analysis, the semantic representation of the menu was evaluated in terms of compactness. Figure 9 shows the accumulated variance after applying PCA to the vectors generated by BERT. The dotted line indicates the threshold of 95% preserved variance, used as a reference to define how many principal components would be retained.

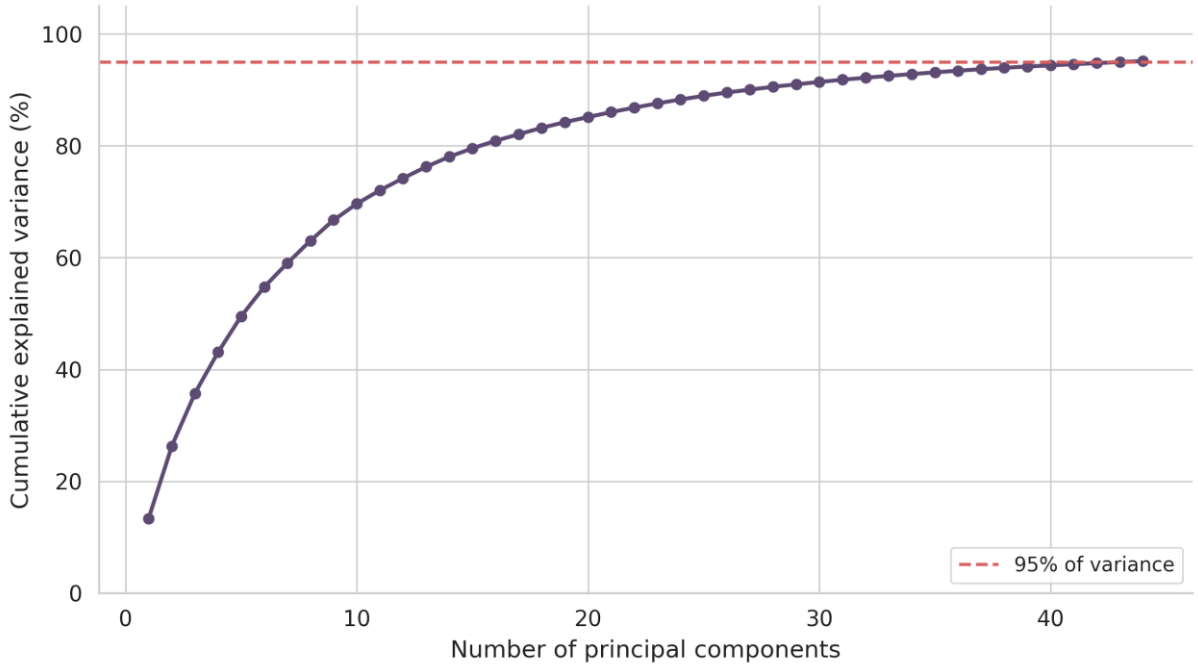



Figure 9. Accumulated variance of BERT components reduced by PCA. Source: authors' own elaboration.

Figure 9 shows that 44 principal components preserve approximately 95.16% of the accumulated variance of the BERT vectors. This means that the textual menu representation could be reduced to a smaller set of variables without discarding most of the statistical information contained in the original vectors. This reduction is relevant because the Gold dataset has 548 records, and using all original BERT dimensions could increase model complexity and the risk of overfitting.

From a conceptual point of view, this stage expands the reading of the menu without replacing manual classification. Manual categories capture explicit groups, such as protein and preparation; BERT vectors can capture semantic proximities between texts. Combining both approaches preserves interpretability and opens a test hypothesis, following Devlin et al. (2019) and Kim et al. (2023), that language representations can extract useful signals from food descriptions.

### 4.10 Consolidation of Gold datasets for modeling

After the exploratory analysis, the resulting artifact is not limited to the visualizations presented. The main technical delivery of this stage is the consolidation of Gold datasets, which organize data at daily granularity and bring together temporal, operational, weather, and menu attributes. These datasets materialize the processing path initiated in Bronze ingestion and qualified in Silver consolidation.

**Table 1. Gold datasets generated for the future predictive modeling stage.**

| Gold dataset | Purpose | Main content | Planned use |
|---|---|---|---|
| **Main tabular Gold** | Gather structured attributes for modeling. | Reservations, historical consumption, lags, moving averages, calendar, weather, and categorized menu. | Train and compare predictive models with interpretable attributes. |
| **Exploratory semantic Gold** | Represent menu text compactly. | Components derived from BERT and reduced by PCA. | Test whether textual menu descriptions add predictive information. |
| **Integrated Gold** | Combine structured and semantic attributes. | Structured attributes enriched with semantic components. | Compare models with and without semantic attributes. |

Source: authors' own elaboration.

Based on this structure, three complementary analytical datasets were generated, each oriented toward different predictive modeling perspectives. The main tabular Gold consolidated 548 records and 140 attributes; the exploratory semantic Gold maintained the same 548 records and 44 components derived from BERT and PCA; and the integrated Gold brought together structured and semantic attributes in an expanded dataset with 184 columns.

These datasets are technically prepared for subsequent predictive modeling stages. Within the scope of this article, the main contribution is to provide traceable, documented analytical datasets that are coherent with the operational problem. Complete predictive validation should be conducted in a specific study, with temporal separation between training and testing, comparison against baselines, and evaluation of error metrics.

### 4.11 Implications of the Gold dataset for future modeling

The Gold datasets indicate that predictive modeling should be conducted by families of attributes, rather than by a single fixed set of variables. Linear models will require attention to scale, collinearity, and the number of indicator variables; tree-based models may explore interactions and expanded categorical variables; and temporal models should treat lags and moving averages carefully to avoid future information leakage.

Another direct implication of the artifact is the separation between observed consumption, estimated consumption, and modeling consumption. This distinction responds to a common limitation in operational datasets: gaps and suspicious records cannot simply be deleted, but they should also not be treated as absolute truth. The developed artifact makes it possible to test different training strategies while maintaining transparency about the criteria used.

Additionally, the integration of BERT and PCA should be understood as an exploratory hypothesis for future evaluation. The presence of 44 semantic components with high preserved

variance indicates dimensional feasibility, but does not, by itself, demonstrate predictive gain. This gain should be measured in a later stage through temporal validation and comparison against baselines with and without semantic attributes.

## 5. CONCLUSION

This article aimed to develop and exploratorily evaluate a Gold analytical dataset intended to support future studies of meal demand forecasting in a school restaurant. To this end, the study began with data ingestion in the Bronze layer, advanced to daily consolidation in the Silver layer, and subsequently structured Gold datasets oriented toward demand analysis and modeling. This path made it possible to transform heterogeneous operational records into more consistent, documented, and traceable analytical datasets.

The results show that the layered organization favored the identification of patterns that would not be evident in the original records. The exploratory analysis indicated that demand is influenced by calendar, recent history, reservations, weekly variability, and menu composition. The visualizations also highlighted the importance of distinguishing observed consumption, estimated consumption, and consumption used for modeling, preventing gaps, suspicious records, or non-operating days from being treated as equivalent observations.

As a practical contribution, the constructed analytical dataset offers structured sets that can support future decisions on production planning, waste reduction, and resource allocation. As a scientific and methodological contribution, the study demonstrates how exploratory data analysis can function as an initial evaluation stage in Design Science Research, connecting data quality, feature engineering, and dataset preparation for predictive modeling.

Some limitations must be acknowledged. The evaluation carried out is exploratory and does not yet compare the performance of predictive models. In addition, the results depend on the completeness and reliability of the available administrative records, especially regarding manual consumption records, reservations, and the menu. There may also be unobserved factors, such as internal events, routine changes, individual student preferences, or specific operating conditions, that influence demand and are not fully represented in the current datasets.

As future work, predictive models should be applied to the constructed Gold datasets using temporal validation and comparison against baselines. It is also suggested to separately evaluate models with structured attributes, models with semantic menu attributes, and models that combine both blocks. Expansion to new periods or other units may also verify the robustness of the analytical dataset and support more sustainable and data-driven solutions for school food management.

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